

Redspher — Full Stack Developer

TypeScript · React · Node.js · AWS · REST · CI/CD
· AI/LLM/MCP

Self-test question list — grouped by topic, no answers. Cover the answer, work through each question aloud, and check yourself against the full guide. **Core** and **Differentiator** tags mark the highest-yield areas.

01 TypeScript — CORE FOCUS

- 1 **interface** vs **type** — when do you reach for each?
 - 2 Explain generics with a real constraint. What does **<T extends ...>** buy you?
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3 `unknown` vs `any` vs `never` ?

4 How do you guarantee a `switch` over a union is exhaustive?

5 What are utility types and which do you actually use?

6 Type guards and discriminated unions?

02 React

— CORE FOCUS

1 `useMemo` vs `useCallback` vs `React.memo` — what problem does each solve, and when is it pointless?

2 What actually triggers a re-render? Why does a component re-render when "nothing changed"?

3 Rules of Hooks — what are they and *why* do they exist?

- 4 The `useEffect` dependency array — behavior for `[]`, `[dep]`, and omitted?

- 5 Controlled vs uncontrolled components?

- 6 Keys in lists — why not the array index?

- 7 How would you debug a slow list / laggy render?

03 Node.js & Backend

— CORE FOCUS

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- 1 Node is single-threaded — how does it handle thousands of concurrent requests?

 - 2 Callbacks → Promises → async/await. What does `async/await` actually do?

 - 3 How do you handle errors in async Express/Node code?
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4 What is middleware and how does the chain work?

5 `process.nextTick` vs `setImmediate`
vs `setTimeout(fn, 0)` ?

04 REST API Design

1 What makes an API RESTful? Design one for an `orders` resource.

2 `PUT` vs `PATCH` , and what is idempotency?

3 Status codes — the ones you'll be expected to place correctly.

4 How do you version and paginate an API?

05 AWS Cloud

- 1 Walk through hosting this app on AWS. Which services, and why?
- 2 Lambda vs EC2 vs Fargate — how do you choose?
- 3 What's a cold start and how do you reduce it?
- 4 S3 + CloudFront for a React SPA — what's the one config people forget?
- 5 IAM in one breath — roles vs users vs policies?

06 Git & CI/CD

- 1 `merge` vs `rebase` — when and why?

2 What actually happens in a CI/CD pipeline?

3 How do you keep secrets and env config out of Git?

07 AI · LLMs

· MCP ·

— DIFFERENTIATOR

Agents

1 What is MCP (Model Context Protocol) and why does it matter?

2 What is an "AI agent" vs a plain LLM call? What does orchestration mean?

3 What is RAG and when do you use it?

4 Function/tool calling — how does it work from a developer's seat?

5 Tokens, context window, temperature — the practical knobs?

- 6** How do you keep LLM features reliable and safe in production?

08 Cross-Cutting & "Translate Business → Technical"

- 1** "Translate a vague business need into a technical solution" — how do you approach it?
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- 2** How do you make a solution "scalable" without over-engineering?
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- 3** PHP legacy is a "plus" — what do you say if asked?
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- 4** Where do you put shared types between a TS frontend and Node backend?